

AI-Generated Characters: Putting Deepfakes to Good Use

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ABSTRACT

AI generated characters, i.e., realistic renderings of human faces, voices, and mannerisms that appear authentic to a human being [7] are made possible through advancements in generative machine learning. In addition to character creation, neural networks have recently also been used for the hyper-realistic synthesis and modification of prose, images, audio, and video data.

While this technology has been most widely associated with media manipulation and the spread of misinformation, often referred to as *deepfakes*, it is increasingly being used for positive applications and integrated into areas ranging from entertainment, to humanitarian efforts and education. With the adaptation and usage of AI generated characters across different industries, we see a potential for significant positive applications in a variety of fields such as learning, privacy, telecommunication, art, and therapy. In this workshop, we will bring together researchers in HCI, AI, and related fields to explore the positive applications, design considerations, and ethical implications of using AI generated characters and related forms of synthetic media.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Interactive systems and tools; HCI design and evaluation methods.**

KEYWORDS

AI generated Characters, Deepfakes, AI, Machine Learning, Generative AI

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1 BACKGROUND

AI-generated characters or infamous *deepfakes* are digital media generated by neural networks, which feature a realistic human appearance and appear authentic to human beings [7]. Such realistic renderings are made possible through advancements in generative machine learning for the hyper-realistic synthesis, recreation, and modification of prose, images, audio, and video data.

While the technology has initially been most widely associated with media manipulation and misinformation, it is also increasingly used for positive applications. For instance, this technology is being integrated into industries ranging from entertainment, like movies or games, to humanitarian efforts and educational tools. One notable example is a recent documentary, “Welcome to Chechnya,”[2] that focuses on a group of activists who risk their lives to fight and

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confront the ongoing anti-LGBTQ fervor in Russia. The documentary team utilized a deepfake-like generative algorithm to protect the identity of anti-LGBTQ movement survivors by overlaying their real faces with the faces of volunteers while preserving their emotions and expressions[4]. Though the technology is far from perfect, it also opens up new opportunities for artists and creators without necessitating Hollywood-sized budgets and timelines. Take for instance, Descript¹, an audio/video editing tool that uses voice-cloning techniques to let users edit or add spoken words to their podcasts or video recordings from simple text. Another example is Dalí Lives,[8] a recent exhibit that uses deepfake technology to recreate Salvador Dalí's face and voice from archival footage to have a version of him interact with museum visitors.

POSITIVE APPLICATIONS OF AI-GENERATED CHARACTERS IN ACTION

With the adaptation and usage of AI-generated characters across industries, we see a potential for future positive applications within a variety of fields such as learning, privacy, tele-communication, art, and therapy. We envision leveraging AI-generated characters to increase motivation and attention amongst learners, enrich interactive role-playing, and create new learning materials. With AI-generated characters, young children could have their teacher take the form of a favorite cartoon or movie character or have historical characters become part of the classroom. In early 2020, this was exemplified by a Chinese professor who saw a spike in student enthusiasm after teaching his online classes as an anime character².

As another application, we see the potential for AI-generated Characters to be used to protect personal identities and enhance communication, both between humans, as well as between humans and digital systems. For example, patients can be de-identified in video recordings without wiping out facial information entirely, such that this data can be used to further research efforts into improving health treatment practices and systems [10].

AI-generated characters with realistic and expressive faces could assume the role of mental health counselors or virtual doctors. These remote counselors would be built by combining the advancements in AI-based conversational agents with the AI-generated character technology. Research shows that socially anxious people reveal more information to virtual counselors [5] and virtual humans. Far less realistic looking characters have been studied for healthcare decision support [3].

Last, we believe AI-generated characters can greatly contribute to the domain of arts, storytelling and culture. Imagine a future in which figures in historical paintings in museums can come to life and speak about their stories, their creation, and about the artists who painted them. Alternatively, portraits of the painters can themselves be used to present the story behind a painting like in the Dalí Lives exhibit. This technique could also be used to preserve world history and cultures and customs that are at risk of disappearing. Similar to how Virtual Reality has been used by holocaust survivors to tell their story[6, 9], AI-generated characters

can support efforts to preserve and archive personal accounts of historical events such that they are not forgotten.

We foresee a future where generative AI will become ubiquitous in our daily lives as part of the ever growing human-AI interaction landscape — but not without challenges [1]. As is the case with most powerful technologies, AI-generated characters are a double edged sword, raising the question of how we ensure safe use of the technology? Our inevitable future with AI generated characters will require us to rethink the fundamentals of human identity, its formation, its safeguarding, and its role in society. In this workshop, we will bring together researchers in HCI, AI and related fields to explore positive applications and discuss design considerations for AI-generated characters and related forms of synthetic media. The goals of the workshop are as follows:

- (1) Develop inspiring and meaningful scenarios, lessons, use cases, and applications for AI-generated characters and other forms of synthetic media.
- (2) Explore opportunities and challenges for designing interactions with AI-generated characters.
- (3) Brainstorm ethical considerations and strategies for the safe deployment of AI-generated characters in society.

2 ORGANIZERS

2.1 Valdemar Danry

Valdemar Danry is a researcher, artist, and tech humanist at the Massachusetts Institute of Technology. He is a member of the MIT Media Lab Fluid Interfaces research group led by Professor Pattie Maes. His passion lies at the intersection of philosophy, technology, and the arts. With his background in psychology, philosophy, and machine learning, he seeks to explore how the augmented body mediates our experiences of ourselves not just as something physical but as something which is "lived". Valdemar's research topics include selfhood, identity, and AI-generated characters, and has been published in conferences such as ACM SIGCHI and ACM Augmented Humans.

2.2 Joanne Leong

Joanne Leong is a PhD Student in the Fluid Interfaces Group at the MIT Media Lab. She has a keen interest in designing technologies that can positively impact how we learn, work, play, and connect with one another. Her research background spans tangible user interfaces, smart textiles, wearable technology, and human augmentation. One of her more recent works explored how manipulating one's self-perception with AI generated media may be used to improve one's communicative and creative abilities. She has completed various internships and academic exchanges in North America, Asia, and Europe, and has published works in top-tier HCI conferences including CHI, UIST (2016 Best Paper) and TEI (2017 Best Paper).

2.3 Pat Pataranutaporn

Pat Pataranutaporn is a multi-disciplinary technologist at the Massachusetts Institute of Technology (MIT). He is a Ph.D. student in the Media Lab's Fluid Interfaces research group. Pat's research lies

¹<https://www.descript.com/>

²https://www.thepaper.cn/newsDetail_forward_6455755

at the intersection of artificial intelligence, human-computer interaction and wearable computing, specifically the area of human-AI symbiosis and cognitive enhancement. Pat has previously worked with global collaborators including NASA TRISH, IBM Research, Bose, ASU, NTU on projects that examine the relationship between people and technology across applications and systems.

2.4 Pulkit Tandon

Pulkit Tandon is a Ph.D. student in the Electrical Engineering department at Stanford University. His research focuses on data compression, neuro-engineering, perceptual engineering and brain-machine-interfaces with components of algorithms, hardware and experiments. He was also an intern at Netflix, where he worked on improving video compression using deep-learning, and on objective video quality metrics. Prior to joining Stanford, he received a bachelor degree in technology from Indian Institute of Technology, Bombay.

2.5 Yimeng Liu

Yimeng Liu is a PhD student in the Perceptual Engineering Lab led by Prof. Misha Sra in the Computer Science department at UCSB. With her background in computer vision, machine learning, human-computer interactions and brain-computer interfaces, Yimeng is enthusiastic about building machine learning enabled applications to enhance human abilities in an interactive and intuitive manner. She has worked on research topics including 3D computer vision, human motion and EEG, and has published papers in ACM UIST.

2.6 Roy Shilkrot

Roy Shilkrot is the chief scientist at Tulip, a MIT Media Lab spinoff. Roy completed his PhD in the MIT Media Lab's Fluid Interfaces research group, where he is currently also a research affiliate with interests in Augmented Reality, Human Computer Interaction, Computer Graphics and Computer Vision.

2.7 Parinya Punpongsanon

Parinya Punpongsanon is an Assistant Professor at the Graduate School of Engineering Science, Osaka University, and a member of the Multidisciplinary Research Laboratory, Osaka University. He received MS and PhD degrees from Osaka University, Japan, in 2013 and 2016, respectively. Prior to his current position, he was a postdoctoral associate at MIT CSAIL from 2017 to 2018. His research interests include human-computer interaction, spatial augmented reality, human perception, and recently personal fabrication.

2.8 Tsachy Weissman

Tsachy's research focuses on Information Theory, Data Compression and Communications, Statistical Signal Processing, Machine Learning, the interplay between them, and their applications, with recent focus on applications to genomic data compression and processing. He is the inventor of several patents and involved in several companies as member of the technical board. A IEEE fellow, he serves on the board of governors of the information theory society as well as the editorial boards of the Transactions on Information Theory and Foundations and Trends in Communications

and Information Theory. He is founding Director of the Stanford Compression Forum.

2.9 Pattie Maes

Pattie Maes is a Professor in MIT's Media Arts and Sciences Program, where until recently she served as Academic Head. Pattie also runs the Media Lab's Fluid Interfaces research group, which aims to radically reinvent the human-machine experience. Coming from a background in artificial intelligence and human-computer interaction, she is particularly interested in the topic of cognitive enhancement or how immersive and wearable systems can actively assist people with memory, attention, learning, decision making, communication, and well-being. Pattie is the editor of three books and serves as an editorial board member and reviewer for numerous professional journals and conferences. Pattie received several awards including an honorary doctorate from the Vrije Universiteit Brussel in Belgium.

2.10 Misha Sra

Sra is the John and Eileen Gerngross Assistant Professor and directs the Perceptual Engineering Lab in the Computer Science department at UCSB. She is affiliated with UCSB's Center for Responsible Machine Learning (CRML), Mind & Machine Intelligence and the Cognitive Science Program. Misha received her PhD at the MIT Media Lab in 2018. She has published at the most selective HCI and VR venues such as CHI, UIST, VRST, and DIS where she received multiple best paper awards and honorable mentions. Her research has received extensive media coverage from leading media outlets (e.g., from Engadget, UploadVR, MIT Tech Review and Forbes India) and has drawn the attention of industry research, such as Samsung and Unity 3D. Her vision is to build technologies that integrate with the brain. Sra's lab works on designing and engineering new AR/VR, hardware and AI technologies that can augment our physiological and perceptual capabilities.

3 PRE-WORKSHOP PLANS

Our website is hosted at: <http://deepfakes.media.mit.edu>.

We will distribute the workshop call through email, social media, and our website. Examples include: ACM SIGCHI mailing lists, MIT mailing lists and social media outlets, Facebook pages and Twitter. We will also reach out to researchers in the area of generative machine learning, synthetic media, and computer vision who may not have a background in HCI but are researching the creative, computational, or ethical aspects of AI-generated characters. The submission portal will be accessed through our website. The paper review timeline is as follows: the call for participation is shared on December 15th, 2021; submission deadline for participation is February 15th, 2022; notification of acceptance happens by February 22th, 2022; and final submissions are due on March 1st, 2022. We will host accepted papers (up to 4 pages long) on the workshop website for participants and others to review prior to the workshop.

4 WORKSHOP STRUCTURE

The one-day workshop will consist of a keynote lecture, research presentations, lively discussion, and group brainstorming. We anticipate 30-60 participants. The workshop will be held online via Zoom

Table 1: Proposed Workshop Schedule

Time	Schedule Item
15 min	Introduction
45 min	Keynote Presentations
15 min	Short Break
45 min	Research Presentations
90 min	Brainstorming
45 min	Lunch Break
45 min	Research Presentations
90 min	Brainstorming
15 min	Short Break
90 min	Next Steps and Ethics Discussion
15 min	Closing

or in-person (depending on the COVID situation). Our website will host a recording of the entire workshop session to support asynchronous viewing for participants that are unable to access the in-person or synchronous virtual meeting.

Authors of accepted papers will be invited to present their papers during the workshop. Following the research presentations, small groups will be assigned to breakout sessions where they will explore specific questions and prompts regarding the opportunities and challenges for designing interactions with AI-generated characters. Participants will work collaboratively to create a visual research map in an online collaborative platform such as "Miro". During the group brainstorm, participants will use post-it notes on Miro to identify opportunities and produce road maps for how this technology may change the future. We will conclude with a reflective discussion on the future and ethics of AI-generated characters and identify directions for further collaboration and discussion. The day's structure is in table 1.

5 POST-WORKSHOP PLANS

We plan to post photos and videos of the workshop on our website and social media during and after the workshop. We will encourage participants who have met through the workshop to submit articles together and comment on other participants' articles on the workshop's website. Our intention is that workshop participants will become co-authors to foster community and collaboration.

6 CALL FOR PARTICIPATION

AI-generated characters (more infamously known as "deepfakes") are a form of digital media generated by deep neural networks, which feature the human appearance and appear authentic to human beings [7]. Such realistic renderings are made possible through advances in generative machine learning for the hyper-realistic synthesis, recreation, and modification of prose, images, audio, and video data.

While the technology is perhaps most widely associated with media manipulation and misinformation, it is increasingly used for positive applications as well. For instance, it is integrated into industries ranging from entertainment, such as movie studios and games, to humanitarian efforts and educational tools. Given the growing interest in AI-generated characters across industries, we

see a potential for future positive applications in a variety of areas including learning, privacy, communication, art, and therapy. This workshop will bring together researchers in HCI, AI, and related fields to explore positive applications and design considerations for AI-generated characters and related forms of synthetic media.

We invite researchers from both academia and industry to submit a short position paper on the theme of AI generated characters and HCI. We will evaluate submissions on fit, ability to stimulate discussion, and contribution to the future of HCI. The workshop's website includes examples of past work in the area to help inspire and inform position papers. Papers should be a maximum of 4 pages, and should be submitted in the CHI Extended Abstracts format. The submission deadline is February 15th, 2022.

At least one author of each accepted position paper must attend the workshop and all participants must register for at least one day of the conference. We will host accepted papers on the workshop's website for participants and others to review prior to the meeting. Submission can be accessed through the website: <http://deepfakes.media.mit.edu/>.

Suggested topics / areas:

- AI-generated media & AI-generated characters
- Case studies of the use of characters in HCI
- Virtual human creation & synthesis pipeline
- Learning with virtual instructors
- Health and wellbeing applications
- Generative Media Synthesis
- Human-AI collaboration
- Authenticity, Style and Improvisation
- AI Embodiment: Can the Mirror Lie?
- Design pitfalls and guidelines
- Ethics of AI-generated characters and their use in HCI

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